

High-speed and Low-Latency Ocular Parallax Rendering Improves Binocular Fusion in Stereoscopic Vision

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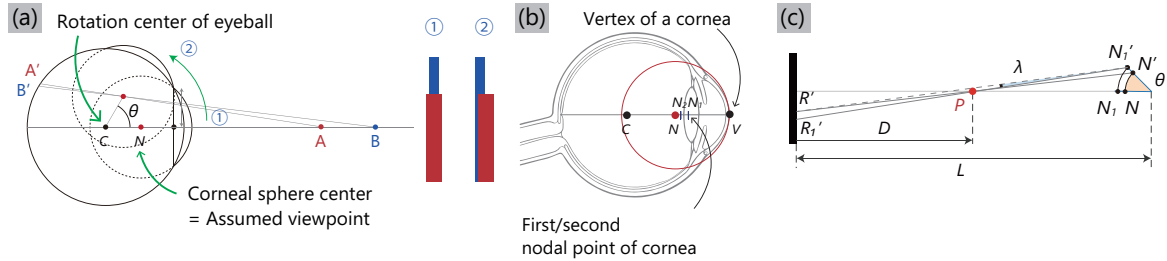


Figure 1: (a) Scene where an apparent gap appears due to ocular parallax. (b) Complex eye model, including primary/secondary nodal points, N_1, N_2 , with a corneal sphere that approximates the corneal anterior surface (red circle) and its center N . (c) Difference of rendered points (R'_1 and R') between ideal and approximated nodal points (N_1 and N).

ABSTRACT

Most of the current virtual/augmented reality (VR/AR) displays use binocular parallax to present 3D images. However, they do not consider ocular parallax, which refers to the slight movement of the viewpoint with eye rotation, such as saccade. A commercial head-mounted display (HMD) has a large latency in realizing ocular parallax rendering. In our study, a high-speed (1,000 fps) and low-latency (average 4.072 ms) ocular-parallax-rendering device was assembled, and its effect was examined wherein a reasonable approximation algorithm for viewpoint was applied. A user study experiment employing a random dot stereogram (RDS) showed improvements in binocular fusion, the causes of which are comprehensively discussed.

Index Terms: Human-centered computing—Interaction paradigms—Mixed / augmented reality; Computing methodologies—Computer vision—Tracking

1 INTRODUCTION

Most of the current virtual/augmented reality (VR/AR) displays produce stereoscopic images using binocular parallax; however, conventional binocular parallax displays ignore the various ocular factors necessary for faithful stereopsis, such as ocular parallax [3] and convergence-accommodation conflict. Displays that respond to these ocular factors are called gaze-contingent displays.

This study focused on ocular parallax, which refers to the parallax produced by a viewpoint shift resulting from eye rotation, as shown in Figure 1 (a). In addition to foveated displays, rendering of ocular parallax can be easily integrated into a VR/AR system using only an eye tracker, which may improve stereoscopic viewing. Currently, head-mounted displays (HMDs) integrate small eye trackers to realize foveated displays [4].

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Previous studies have examined the effects of ocular parallax rendering on off-the-shelf HMDs with built-in eye trackers [2]. However, these systems initially have a large latency of nearly 20 ms, which may, to a certain extent, degrade the quality of stereoscopic vision, such as perceived depth.

In this study, a high-speed and low-latency ocular parallax rendering system is proposed to solve the aforementioned limitations. To ensure the speed and stability of eye nodal point tracking, an approximation method of corneal nodal position is also proposed and integrated into our system. As a first step for studying stereoscopic vision using this system, an experiment on binocular fusion was performed.

2 METHOD

In the eye, the viewpoint is provided by the primary nodal point of the cornea N_1 , as shown in Figure 1 (b). The eye tracking algorithm used is the pupil-centered corneal reflection (PCCR) method, where multiple corneal reflections of infrared light are used to estimate a gaze accurately and quickly [1]. From the perspective of eye tracking algorithms, tracking the correct position of the primary nodal point initially tends to be inaccurate and unstable because of the difficulty of the elliptic approximation required for pupil shape, especially in high-speed vision. A dark pupil method, where only one pupil appears black while the other does not through irradiation by infrared light is used to detect the pupil, which usually has insufficient contrast in high-speed vision.

Using this algorithm, the corneal sphere center N is calculated. Its position is close to the primary nodal point N_1 , as shown in Figure 1 (b). For example, as a typical example, the distance between N_1 and the vertex of cornea V is within 6.5-7.0 mm while the typical value between N and V is 7.8 mm [2]; note that those values vary individually. The difference in the rendering position is simulated, as shown in Figure 1 (c), which proved to be trivial based on the author's simulation. Therefore, this study tracks N instead of N_1 because the algorithm is simpler and the tracking accuracy, speed, and stability are higher.

3 EXPERIMENTS

The apparatus used in the user study is displayed at the top of Figure 2. For eye tracking cameras, two Ximea MQ013RG-ON and two Basler acA800-510um, were used. Both were monochrome,

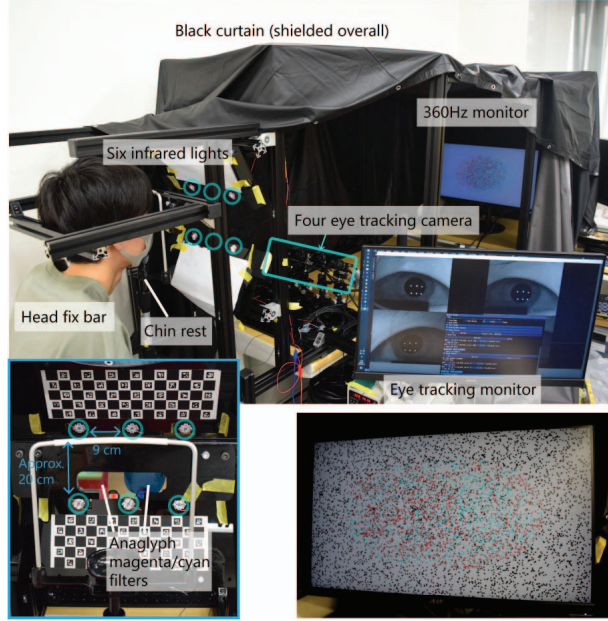


Figure 2: Top: experimental apparatus. Bottom: the left and right images show RDS with or without background random dots, respectively.

and the ROI was reduced to 512×410 px to enable an acquisition rate of 1000 fps. The display was an LCD gaming monitor (Acer XV252QFbmiiprx, 1920×1080 px, 360 Hz standard, 390 Hz overlocked; latency of GTG (gray to gray): 1 ms standard, 0.5 ms overdrive at minimum) which was set 2.5 m away from the user position and was overlocked as well as overdriven for maximum performance. GALLERIA ZA9C-R38 (Windows 10 Pro; CPU: Intel Core-i9 12900K; GPU: GeForce RTX 3080; RAM: 64GB; SSD 2TB) was used.

The frequency and response latency of the display were evaluated using Photron FASTCAM SA-X2 (20k fps) as 359.827 ± 109.232 fps and 1.554 ± 0.440 ms, respectively. Assuming that the camera operates at 1,000 fps exactly, the total system latency can be estimated as 4.072 ± 0.2886 ms, which is regarded as sufficiently small.

3.1 User study on binocular fusion

The stereoscopic image was presented using an elliptic RDS with a random dot background, as shown in the bottom left of Figure 2. There were nine disparities in presenting the stereoscopic images: -0.75° , -0.5° , -0.25° , 0.5° , 1.5° , 2.5° , 3.5° , 4.5° , and 5.5° . Eighteen different experimental conditions were used, with and without (w/o) ocular parallax rendering. There were five trials for each condition, and therefore, $5 \times 18 = 90$ trials in total. Regarding the trial order, to ensure that the order was not biased, five sets with 18 conditions each, were shuffled and concatenated. Twelve subjects aged 21–35 years, with good vision at a distance of 2.5 m with the naked eye or contact lenses, were recruited by snowball sampling. All were Japanese, except for one mixed-race Thai-Japanese woman and one Indian woman; the male to female ratio was 1:1, indicating equal participation. All patients had black or brown eyes.

The process per trial was as follows. The subject saw an RDS image for up to 5 s and immediately pressed a specified key when he/she perceived the ellipse image floating, as instructed beforehand. If the key was pressed, the presentation continued for an additional 2 s and then disappeared. Otherwise, it disappeared immediately after 5 s of presentation. Subsequently, the screen blacked out; the infrared lights were off; and a 3-s break followed.

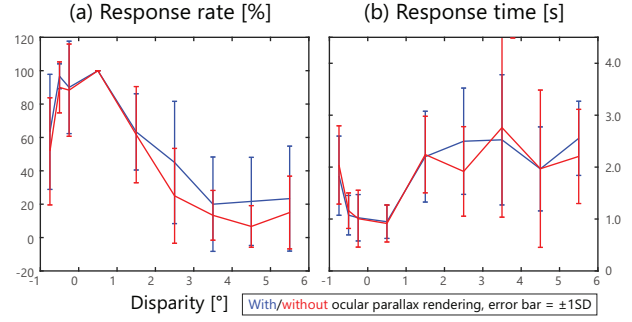


Figure 3: Statistical results of binocular fusion for all subjects, showing (a) the response rate and (b) the response time of when the subjects answered.

The statistical results are shown in Figure 3, where the response rate gradually decreased as the absolute disparity increased. The response rate was higher with our ocular parallax rendering under the conditions where disparity was equal to or greater than 2.5° while the response time was not.

The generalized linear mixed model (GLMM) analysis for each response rate and time, focusing on the factor of w/o ocular parallax rendering was $\chi^2(8) = 21.02$, $p = 1.255E - 2$ and $F(1, 587) = 1.7185$, $p = 0.2938$, respectively, where only the former showed a statistically significant difference. By contrast, the interaction of the response rate was $\chi^2(8) = 5.359$, $p = 0.7186$ without a statistically significant difference.

Therefore, it can be said that binocular fusion in stereoscopic vision was improved with high-speed ocular parallax rendering.

4 CONCLUSION

A high-speed and low-latency stereoscopic viewing device was assembled to accurately reproduce the ocular parallax, which integrated a reasonable approximation of the tracked viewpoint, that is, the nodal point of the cornea. Further, the effect of the device on binocular fusion in stereoscopic vision was investigated in a user study, where improvements were observed with ocular parallax rendering.

In a future study, comparative evaluation with an HMD-based study [2] should be performed considering multiple factors of stereoscopic vision, including perceived depth.

ACKNOWLEDGMENTS

This work was approved by Life Science Research Ethics and Safety, the University of Tokyo and supported by JST, ACT-X Grant Number JPMJAX21KH, Japan. We thank Taiki Fukiage, NTT Communication Science Laboratory, for useful discussions, and Ishikawa Group Laboratory, the University of Tokyo, for equipment rental.

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